

AI LEARNING PRODUCT

Final Product Strategy & Discussion Overview

From document upload to a personalised AI tutor experience

Core direction: Manually prove the ideal learning experience first, establish a quality benchmark, then reverse-engineer and automate the proven process.

Product pillars: 15 / 30 / 45-minute lessons • Personalisation • Engaging narration • IQ / EQ / SQ / Research Skills • Cognitive Engagement Score (CES)

1. Executive Summary

The product should become a personalised AI tutor that helps users learn from PDFs, textbooks, books, and other learning documents. It should understand both the content and the learner, then generate a structured lesson with slides, narration, audio, questions, examples, and skill-building activities.

The key product decision: do not automate the entire lesson-generation pipeline first. Create excellent lessons manually, test them with real users, define the quality threshold, learn which prompts and models work, and then automate the proven workflow.

The experience should move from simple content delivery to a tutor that knows the learner's goal, adapts the lesson to the selected duration, uses engaging narration, develops IQ, EQ, SQ and research skills, and measures meaningful engagement through CES.

2. Product Vision

The user uploads a document and receives a learning experience tailored to their goal, level, available time, and chapter-specific needs.

- Understand the learner and their long-term goal.
- Understand the uploaded book and its learning context.
- Understand the current chapter and optional user instructions.
- Offer 15-minute, 30-minute, and 45-minute lesson formats.
- Explain technical concepts accurately and simply.
- Use real-world examples and questions to build understanding.
- Make narration feel like a skilled, engaging tutor rather than a text-to-speech reader.
- Map learning activities to IQ, EQ, SQ, and research skill.
- Measure meaningful engagement and growth through CES.

3. Proposed Learning Formats

15 minutes	30 minutes	45 minutes
Purpose Fast, focused understanding and revision.	Purpose Balanced learning and revision.	Purpose Deep, immersive concept building.
Key concepts, definitions, formulas, 1-2 examples, quick recall.	Important concepts, examples, questions, connections, checkpoints.	Detailed teaching, examples, interaction, personalisation, skill activities, revision.

Important: duration should change more than slide count. It should change depth, pace, examples, questions, personalisation, and skill-building.

4. Three Forms in the Application

The product should collect information at three different levels so users do not have to repeat the same information everywhere.

4.1 User Onboarding Form

- Main learning goal and reason for using the product.
- Exam, interview, project, job, academic, or personal-learning purpose.
- Current knowledge level and difficult areas.
- Preferred language, teaching style, tone, examples, and pace.

- Typical available learning time and motivation.
- Career or academic goal.

4.2 Book Understanding Form

- Why the user uploaded the book or PDF.
- What they want to achieve from it.
- Complete book vs selected chapters.
- Important or difficult sections.
- Deadline and expected depth.
- Whether the document should be followed exactly or reorganised for learning.

4.3 Chapter / Sub-Chapter Form

- What the user wants to understand from this chapter.
- Specific doubts or difficult topics.
- Desired duration and depth.
- Need for examples, formulas, diagrams, or questions.
- Topics to focus on or skip.
- Expected outcome by the end of the lesson.

The chapter form should be short and optional. Existing profile and book information should not be repeated.

5. User Prompt + System Prompt

User prompt: an optional, direct instruction from the learner about what they want from the current lesson. Examples include focusing on exam-important concepts, explaining formulas step by step, using a particular type of example, or asking questions after each topic.

System prompt: the internal product instruction that controls how the lesson should be generated. It combines the uploaded content, user profile, book context, chapter context, user prompt, duration, past performance, skill mapping, and teaching rules.

Recommended priority: accuracy and safety → source content → system teaching rules → duration → user profile → book context → chapter instructions → user prompt → past performance.

6. Manual Testing Before Automation

This is the central development strategy.

Stage	What happens
1. Manual experiment	Create a complete lesson manually using AI models for structure, slides, narration, and voice.
2. Quality benchmark	Review accuracy, coverage, flow, engagement, narration, duration, personalisation, and skill development.
3. Prompt refinement	Identify which prompts, models, structures, and instructions produce the best output.
4. Reverse engineer	Document the successful process and every manual correction required.
5. Automate	Build the proven process into the product and compare automated output against the manual benchmark.

Recommended first test: build one complete 45-minute lesson. It exposes the largest number of product requirements. Once stable, adapt the proven approach to 30 and 15 minutes.

7. Narration Strategy

Narration should not simply read the slide. It should behave like an intelligent, engaging, personalised tutor.

- 60% technical teaching: accurate concepts, definitions, formulas, mechanisms, applications, and explanations.
- 40% engagement: humour, light satire, friendly roasting, stories, encouragement, questions, motivation, and goal reminders.
- Humour must support attention and memory. It should never reduce technical accuracy or make the learner uncomfortable.
- The narrator should add value beyond what is visible on the slide.

8. Teaching Flow

- 1. Introduce: what are we learning and why does it matter?
- 2. Explain technically: give the correct definition, rule, process, formula, or mechanism.
- 3. Simplify: explain the same idea in easy language.
- 4. Apply: connect it to a real-world or user-relevant example.
- 5. Ask: make the learner predict, recall, compare, solve, or explain.
- 6. Reflect: allow a short thinking or response moment.
- 7. Correct: explain the answer and why it is correct.
- 8. Connect: link the concept to previous and upcoming content.
- 9. Reinforce: connect the concept back to the learner's goal.

9. Personalisation

Level	Source	Used for
User-level	Onboarding profile	Long-term goals, background, level, language, motivation, preferences
Book-level	Book form	Purpose, important chapters, deadline, expected depth
Chapter-level	Chapter form + user prompt	Immediate doubts, duration, focus areas, examples, desired outcome

The tutor should use personalisation naturally throughout the lesson, not only in the introduction. It can remind the learner how a concept supports their exam, interview, project, career, or other stated goal.

10. IQ, EQ, SQ and Research Skill

The product should not only transfer information. It should deliberately develop thinking and learning skills.

Skill	Focus	Example activities
IQ	Logic, analysis, recall, problem-solving, pattern recognition, application	Solve, predict, compare, calculate, connect concepts
EQ	Confidence, motivation, self-awareness, persistence, response to mistakes	Confidence checks, reflection, recovery after errors
SQ	Communication, perspective-taking, collaboration, explanation	Teach-back, interview simulation, explain to another person

Skill	Focus	Example activities
Research	Questioning, evidence, source comparison, verification, assumptions	Identify missing evidence, verify claims, form research questions

11. Cognitive Engagement Score (CES)

IQ, EQ, SQ, and research skill form a core product functionality called the Cognitive Engagement Score (CES). CES should measure how actively and meaningfully the learner engages with the lesson, not simply how long they watched it.

Potential signals include answer accuracy, attempts, confidence, response to hints, ability to explain concepts, application ability, questions asked, persistence after mistakes, recall, and research behaviours.

Initial weighting hypothesis for testing: IQ 35% + EQ 25% + SQ 20% + Research 20%. This is a product hypothesis, not a validated final formula.

Example: $CES = (IQ \times 0.35) + (EQ \times 0.25) + (SQ \times 0.20) + (Research \times 0.20)$.

The weights should adapt to the user's goal when evidence supports doing so. For example, exam preparation may need more IQ emphasis, while interview preparation may increase SQ and EQ.

CES should be framed as a growth measure. Avoid labels such as "low IQ." Use language such as "concept application is improving" or "evidence-checking needs more practice."

12. CES as an Adaptive System

- Low IQ engagement → simpler explanations, more worked examples, more practice.
- Low EQ engagement → shorter sections, encouragement, confidence-building activities.
- Low SQ engagement → more communication and explanation tasks.
- Low research engagement → more evidence-checking and questioning tasks.
- Strong performance → deeper questions and more advanced applications.
- High confidence with low accuracy → reflection and correction.
- Performance and engagement should influence future lessons.

13. Skill Mapping by Lesson Duration

15 min	30 min	45 min
Essential IQ checks, one reflection or confidence question, one short application.	Multiple IQ questions, EQ checkpoints, one SQ activity, one research-thinking activity.	Full skill mapping, multiple cognitive activities, emotional reflection, communication practice, research-based questioning, stronger CES measurement.

14. Slides + Narration

Slides	Narration
Main concepts, definitions, formulas, diagrams, key steps, short examples, questions, visual summaries.	Detailed explanation, context, simplification, stories, examples, humour, transitions, questions, motivation, personalisation.

The slide is the visual guide. The narration is the teacher. The narrator should not read every bullet word for word.

15. End-to-End Product Data Flow

Input: user profile → uploaded book → book context → selected chapter → chapter context → user prompt → duration → learning history → previous CES.

Processing: document understanding → chapter segmentation → concept detection → difficulty estimation → goal matching → lesson planning → skill mapping → slide generation → narration generation → questions → personalisation → duration control → audio generation.

Output: slides → narration → audio → questions → activities → personalised reminders → chapter summary → CES → skill insights → next-lesson recommendation.

16. Quality Threshold

Area	What to check
Accuracy	Is the teaching technically correct?
Source alignment	Does it correctly represent the uploaded material?
Coverage	Are important concepts included?
Duration	Does it match the selected time?
Structure	Does the lecture follow a logical flow?
Narration	Does narration add value beyond the slide?
Engagement	Does it make the learner think and participate?
Personalisation	Does it use the learner's goals and context naturally?
Skill development	Are IQ, EQ, SQ and research skills meaningfully mapped?
CES	Can engagement be measured consistently and fairly?

17. Manual Experiment Documentation

- User profile, book, chapter, user prompt, duration.
- Lesson outline, slide prompt, narration prompt, question prompt, skill-mapping prompt.
- AI model, voice model, audio speed, final duration.
- IQ, EQ, SQ and research activities included.
- CES signals collected.
- Manual corrections required and problems found.
- Learner feedback and final quality score.

18. Immediate Next Actions

Phase	Actions
Define	Finalise forms, duration standards, prompts, skill indicators and CES hypothesis.
Manually produce	Create one complete 45-minute lesson with slides, narration, audio, questions, activities and CES signals.
Evaluate	Measure accuracy, understanding, engagement, duration, narration, personalisation and CES.
Refine	Improve prompts, lesson structure, narration, forms, skill mapping and CES.

Phase	Actions
Automate	Convert the proven manual workflow into product workflows and compare against the benchmark.

19. Final Product Principle

The team should not begin by asking: “How can we automatically generate a lecture?”

The better question is: “What does an excellent, personalised, engaging and measurable 15-minute, 30-minute or 45-minute learning experience look like?”

Once the team can repeatedly create that experience manually, it can build the technology to reproduce it reliably.

20. Final Product Direction

- Understands the learner through onboarding.
- Understands the uploaded book and chapter.
- Accepts optional chapter-specific instructions and a user prompt.
- Uses a strong internal system prompt.
- Offers 15-, 30-, and 45-minute lessons.
- Generates slides and narration as one connected experience.
- Explains technical concepts accurately and simply.
- Uses real-world examples and interactive questions.
- Uses humour and human connection carefully.
- Connects learning back to the user's goals.
- Develops IQ, EQ, SQ and research skill.
- Measures meaningful engagement through CES.
- Uses past performance to improve future lessons.
- Helps users understand, apply, explain and retain knowledge.

Bottom line: Prove the learning experience manually first. Establish the benchmark. Reverse-engineer what works. Then automate it.